

RAG (Retrieval Augmented Generation) - Complete Notes

RAG allows an LLM to answer questions using external data.

Core Components

1. LLM - Generates final answers
2. Tokens - Small pieces of text
3. Embeddings - Numerical representation of meaning
4. Vectors - Lists of numbers representing text
5. Chunks - Small sections of documents
6. Vector Database - Stores vectors and performs similarity search

RAG Pipeline

PDF -> Loader -> Chunking -> Embeddings -> Vector Database

Question -> Embedding -> Similarity Search -> Top Chunks -> LLM -> Answer

Practical Steps

1. Load PDF
2. Create chunks
3. Generate embeddings
4. Store vectors in ChromaDB/Pinecone
5. Convert question to embedding
6. Retrieve top-k chunks
7. Send context to LLM
8. Generate final answer

Interview One-Liner

RAG retrieves relevant information from a vector database using embeddings and similarity search, then provides that information to an LLM to generate accurate answers.